Matplotlib

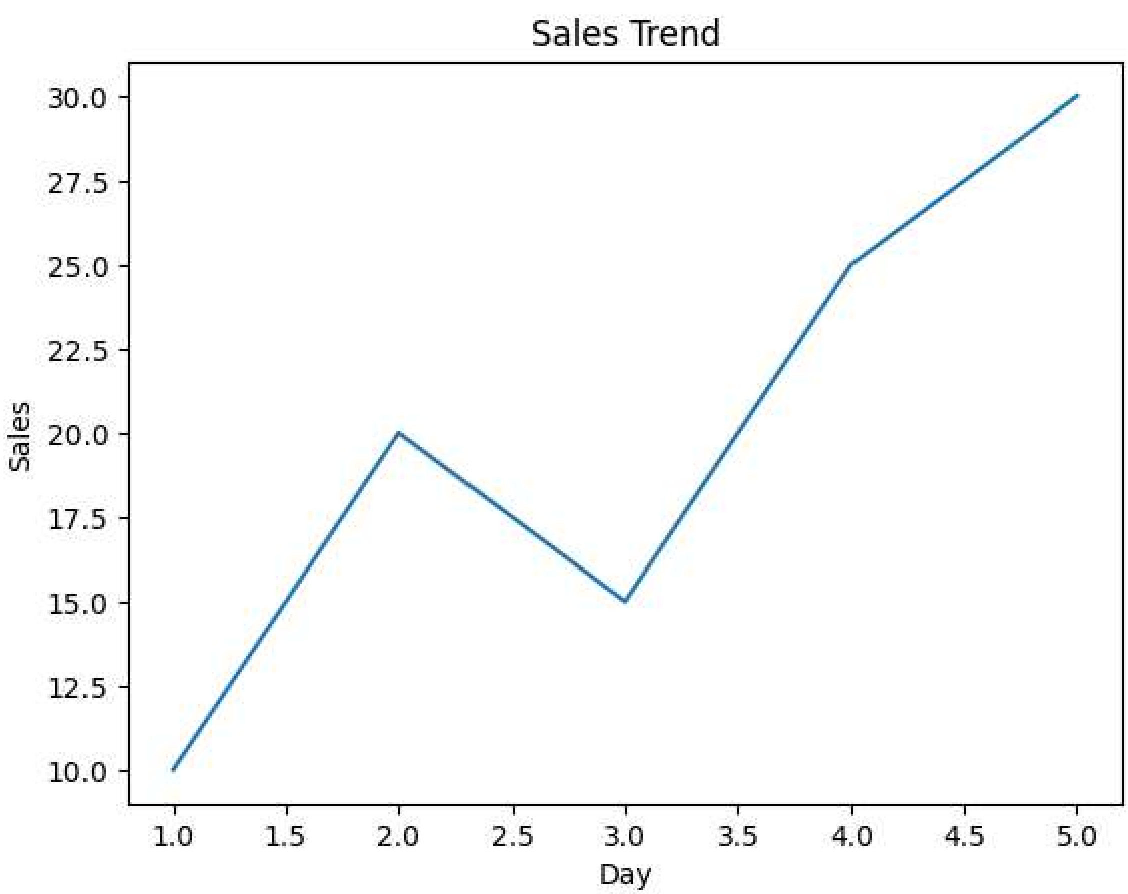
- Used for data visualization

1: Line Plot (Trend ke liye)

```
Kab use karein?
```

- Time series data
- Growth ya decline dekhna ho

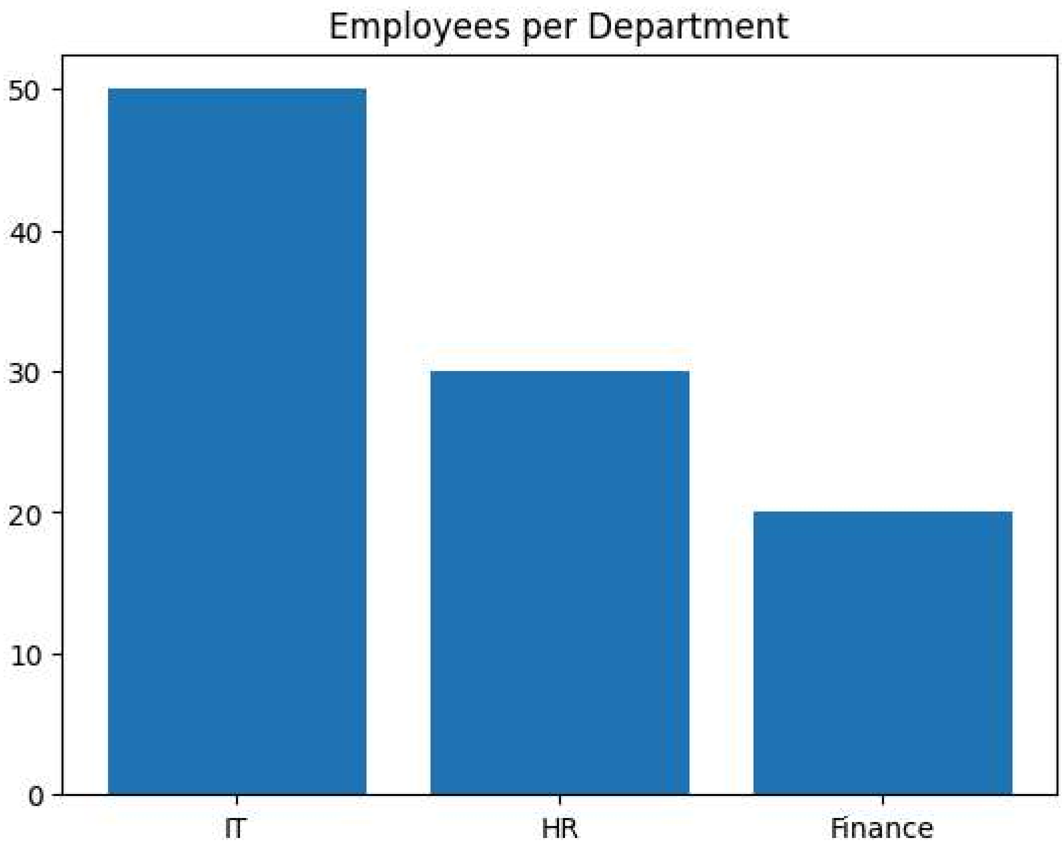
In [1]:
```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,15,25,30]

plt.plot(x, y)
plt.xlabel("Day")
plt.ylabel("Sales")
plt.title("Sales Trend")
plt.show()
```



2: Bar Plot (Category Comparison) Kab use karein?

- Different categories compare karni ho

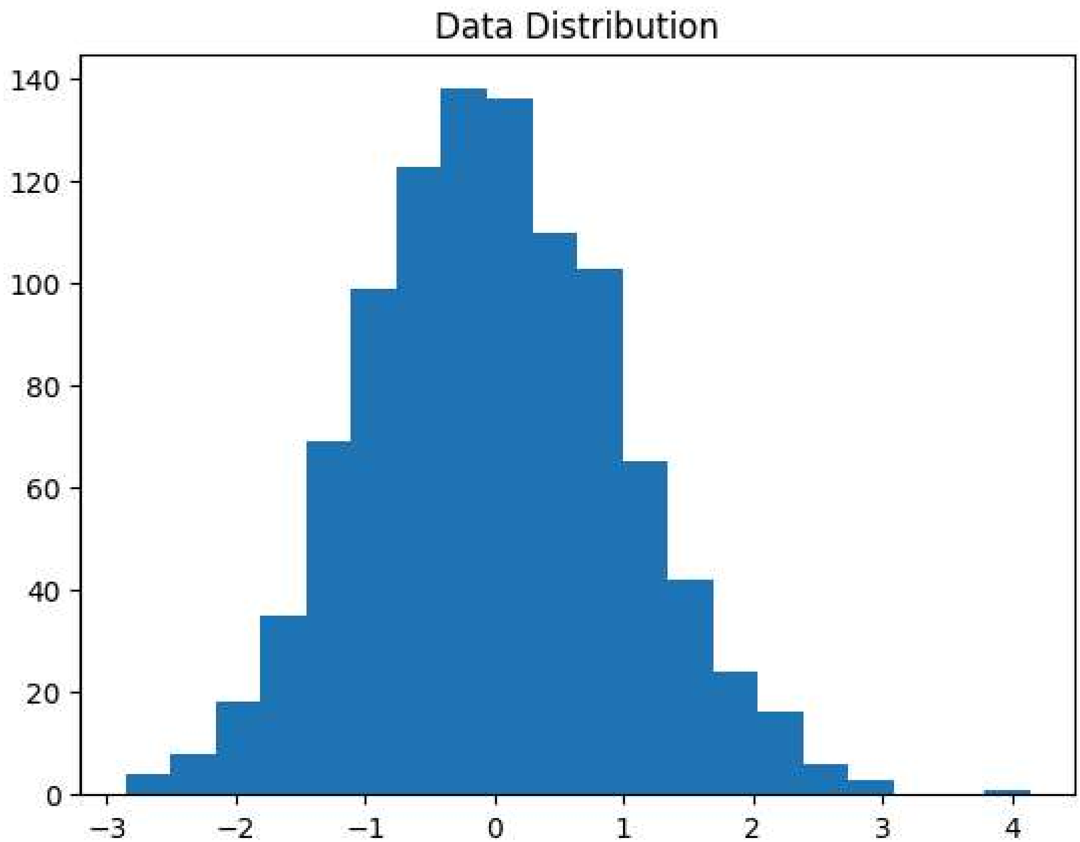
In [2]:
```
categories = ["IT","HR","Finance"]
values = [50,30,20]

plt.bar(categories, values)
plt.title("Employees per Department")
plt.show()
```



3: Histogram (Distribution dekhne ke liye) Kab use karein?

- Continuous data ka spread dekhna ho

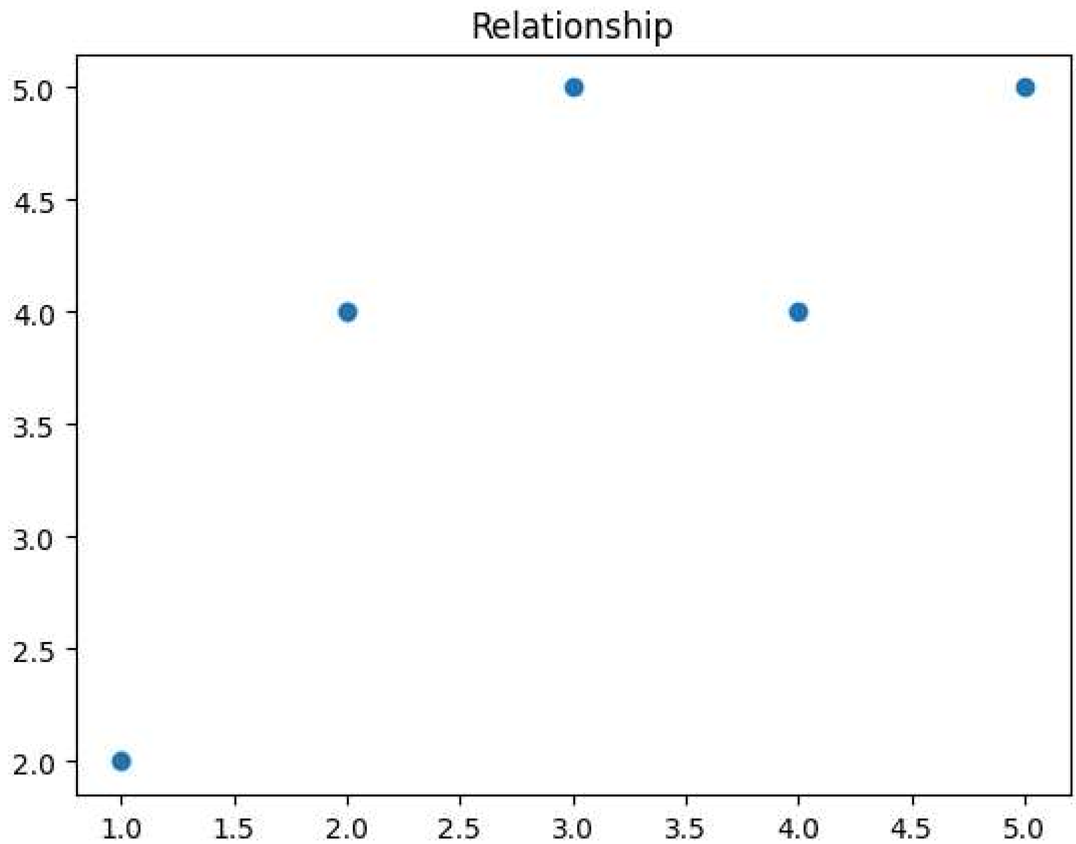
In [4]:
```
import numpy as np

data = np.random.randn(1000)

plt.hist(data, bins=20)
plt.title("Data Distribution")
plt.show()
```



4: Scatter Plot (Relationship Check) Kab use karein?

- 2 variables ka relation dekhna ho

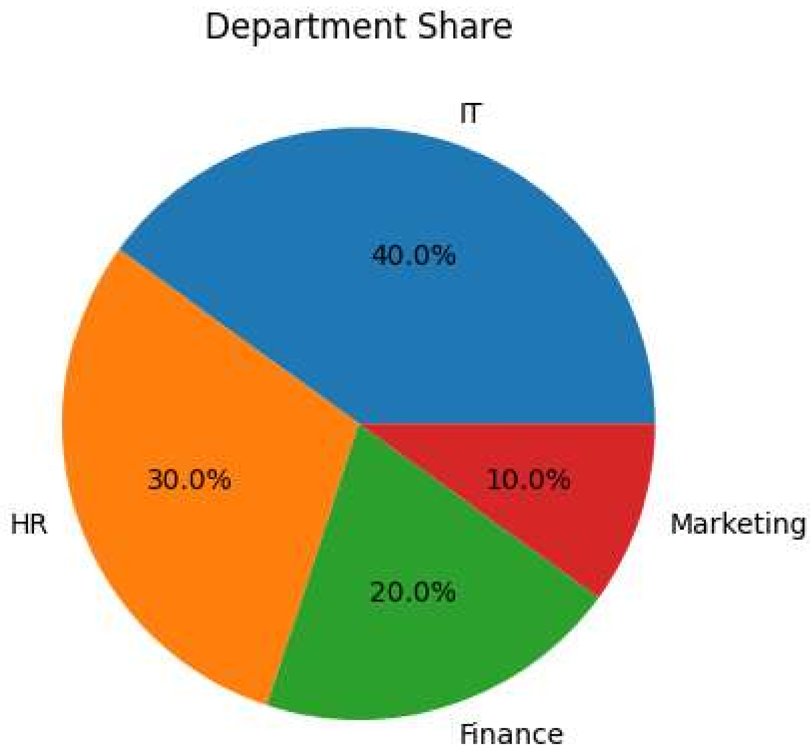
In [5]:
```
x = [1,2,3,4,5]
y = [2,4,5,4,5]

plt.scatter(x, y)
plt.title("Relationship")
plt.show()
```



5: Pie Chart (Percentage Share) Kab use karein?

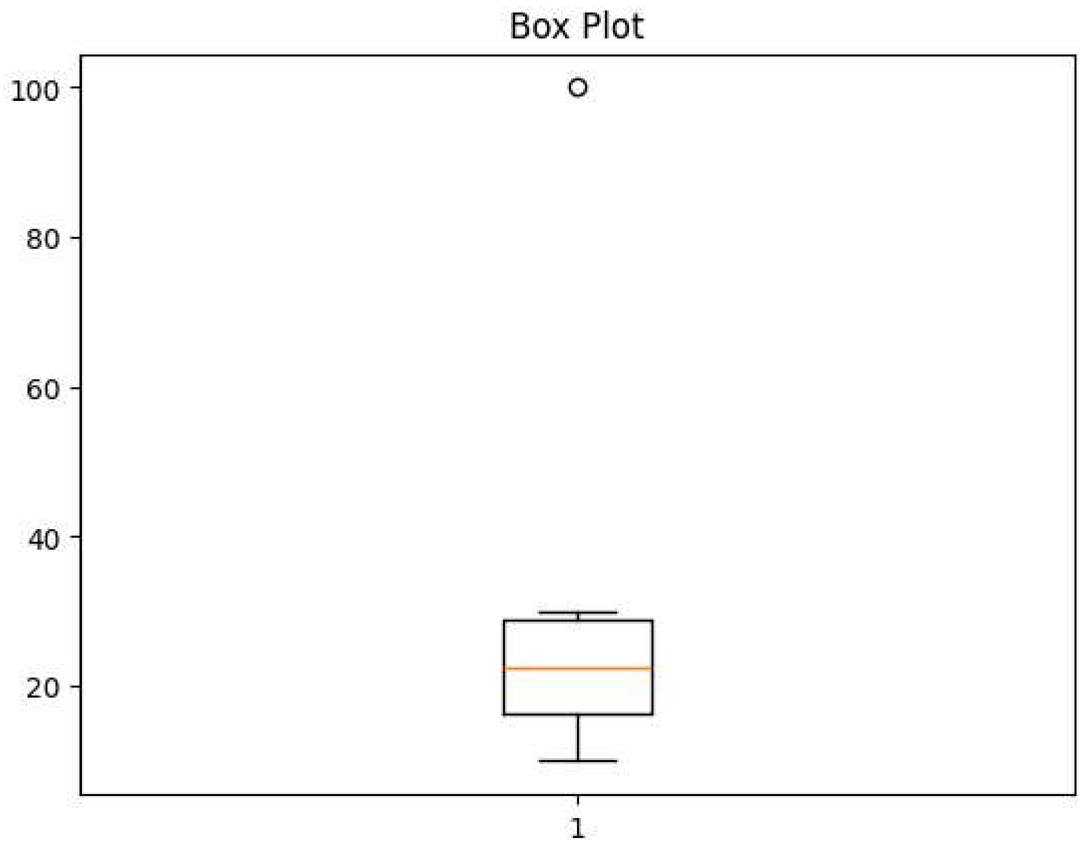
- Part-to-whole relationship

In [7]:
```
sizes = [40,30,20,10]
labels = ["IT","HR","Finance","Marketing"]

plt.pie(sizes, labels=labels, autopct="%1.1f%%")
plt.title("Department Share")
plt.show()
```



6: Box Plot (Outliers Check)

In [6]:
```
data = [10,20,15,25,30,100]

plt.boxplot(data)
plt.title("Box Plot")
plt.show()
```

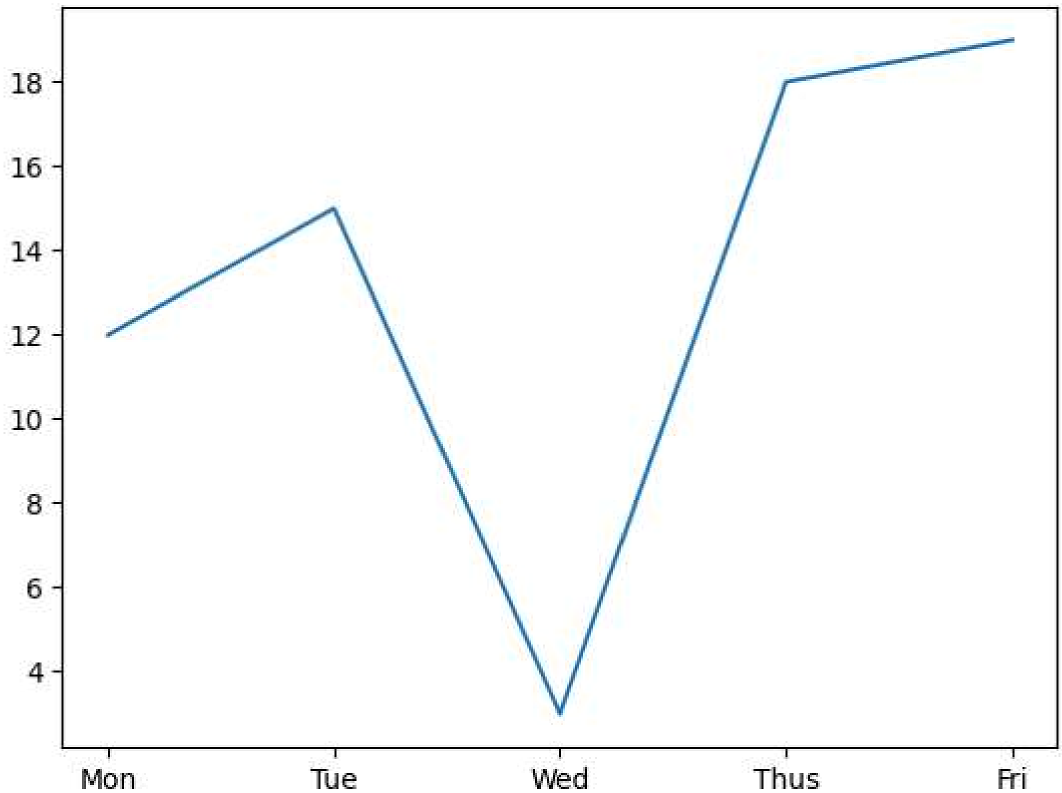


Key Terms You Should Know

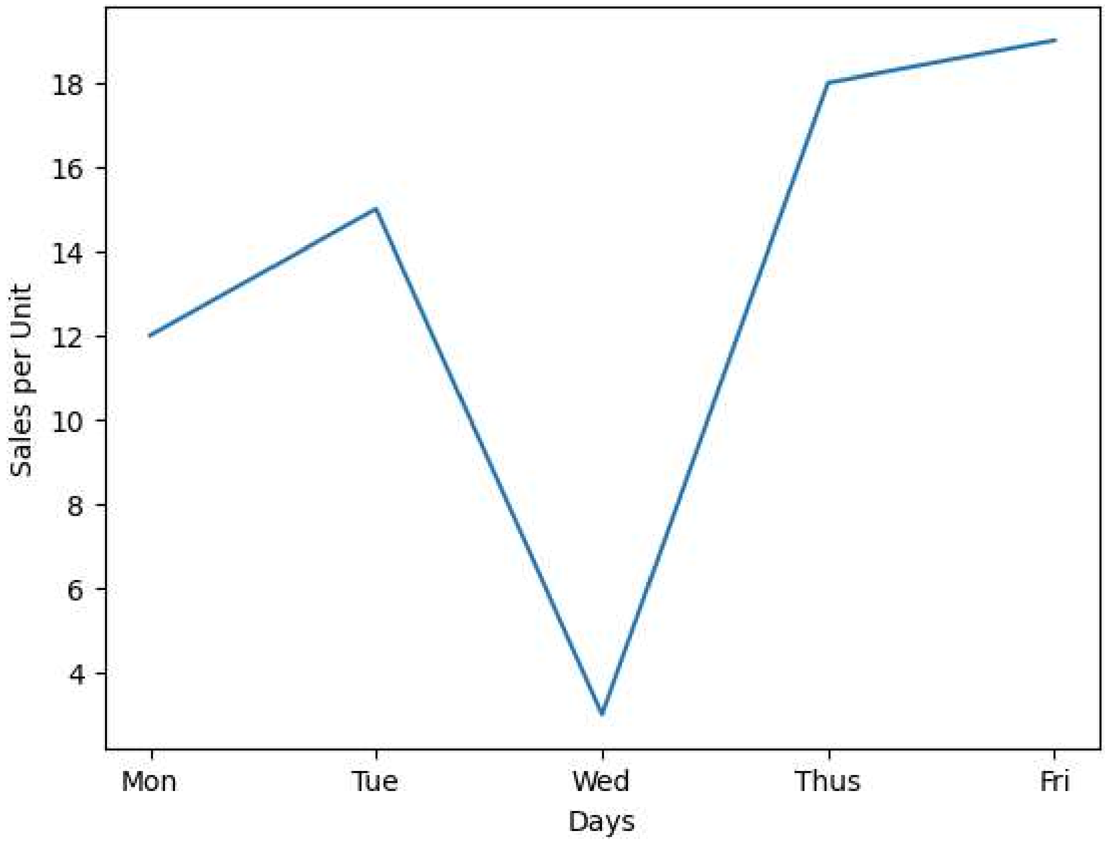
- Data point
- x-axis / y-axis
- Figure (Whole Canvas)
- Axes (Also Called as Graph Box)
- Plot
- Marker
- Line Style
- Color
- Legend (Kon see line kya represent kr rahi hy)
- Label
- Title
- Grid
- Function
- Method
- Parameters
- Keyword Arguments
- Object-Oriented API (Advance way of plotting the graphs)
- DPI (Dots per inch)(jo graph hum banaye ga usko kis resolution ma save krna hy)
- Backend (tkagg(default),webagg,qt5agg)

In [12]:
```
import matplotlib.pyplot as plt
x=["Mon","Tue","Wed","Thus","Fri"]
y=[12,15,3,18,19]
plt.plot(x,y)
```

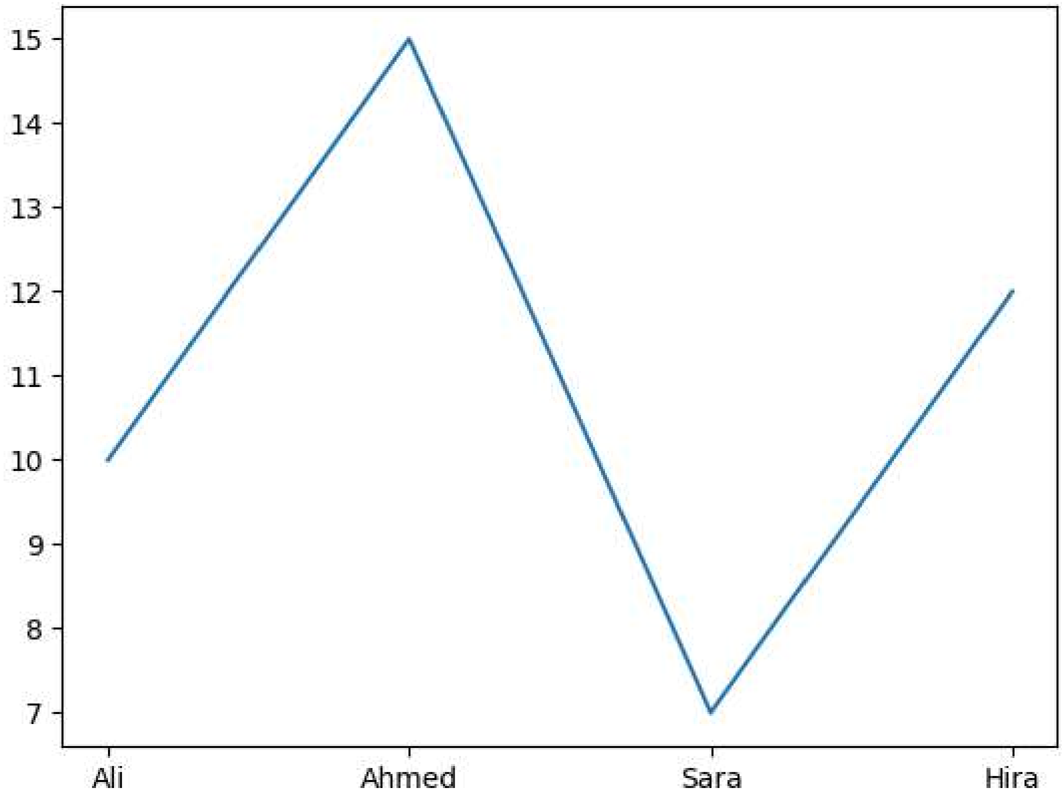
Out[12]:
```
[<matplotlib.lines.Line2D at 0x1a38eef2cc0>]
```



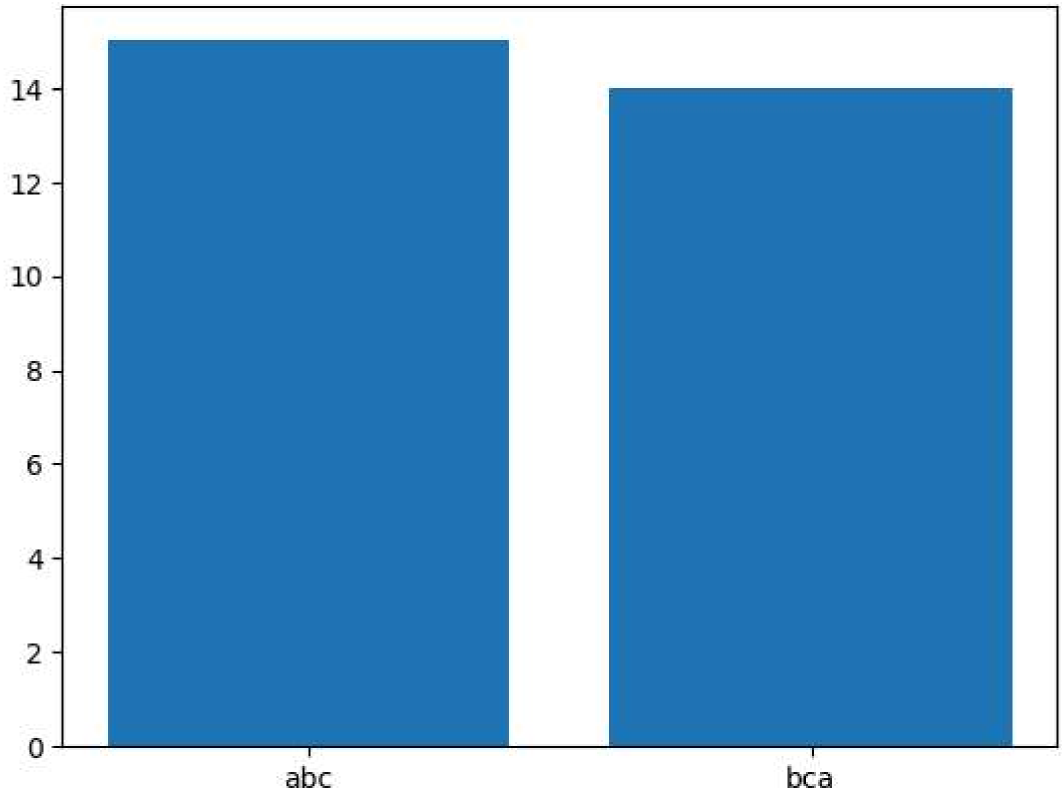
In [10]:
```
import matplotlib.pyplot as plt
import pandas as pd
data = {
    "Days":["Mon","Tue","Wed","Thus","Fri"],
    "Sales":[12,15,3,18,19]
}
df = pd.DataFrame(data)
plt.plot(df["Days"],df["Sales"])
plt.xlabel("Days")
plt.ylabel("Sales per Unit")
plt.show()
```



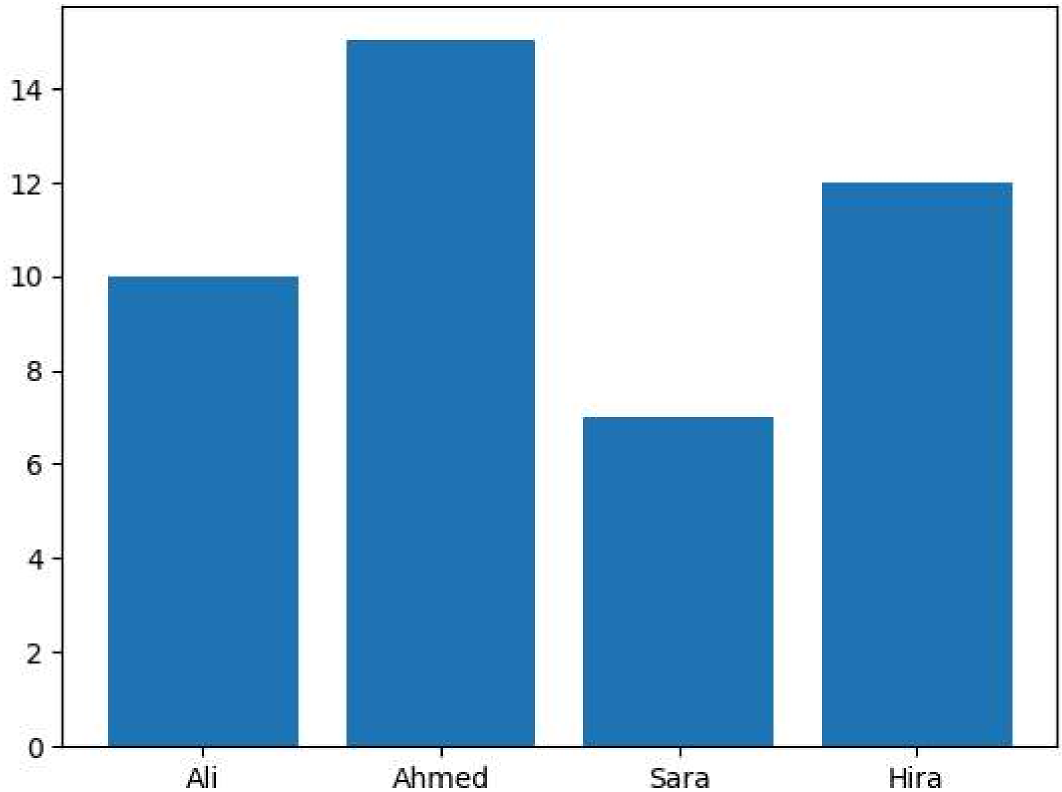
In [16]:
```
import matplotlib.pyplot as plt

data = {
    "Ali": 10,
    "Ahmed": 15,
    "Sara": 7,
    "Hira": 12
}

x = list(data.keys())
y = list(data.values())

plt.plot(x, y)
plt.show()
```



In [14]:
```
import matplotlib.pyplot as plt

data = {}

n = int(input("Kitne students: "))

for i in range(n):
    name = input("Name: ")
    marks = int(input("Marks: "))
    data[name] = marks

x = list(data.keys())
y = list(data.values())

plt.bar(x, y)
plt.show()
```



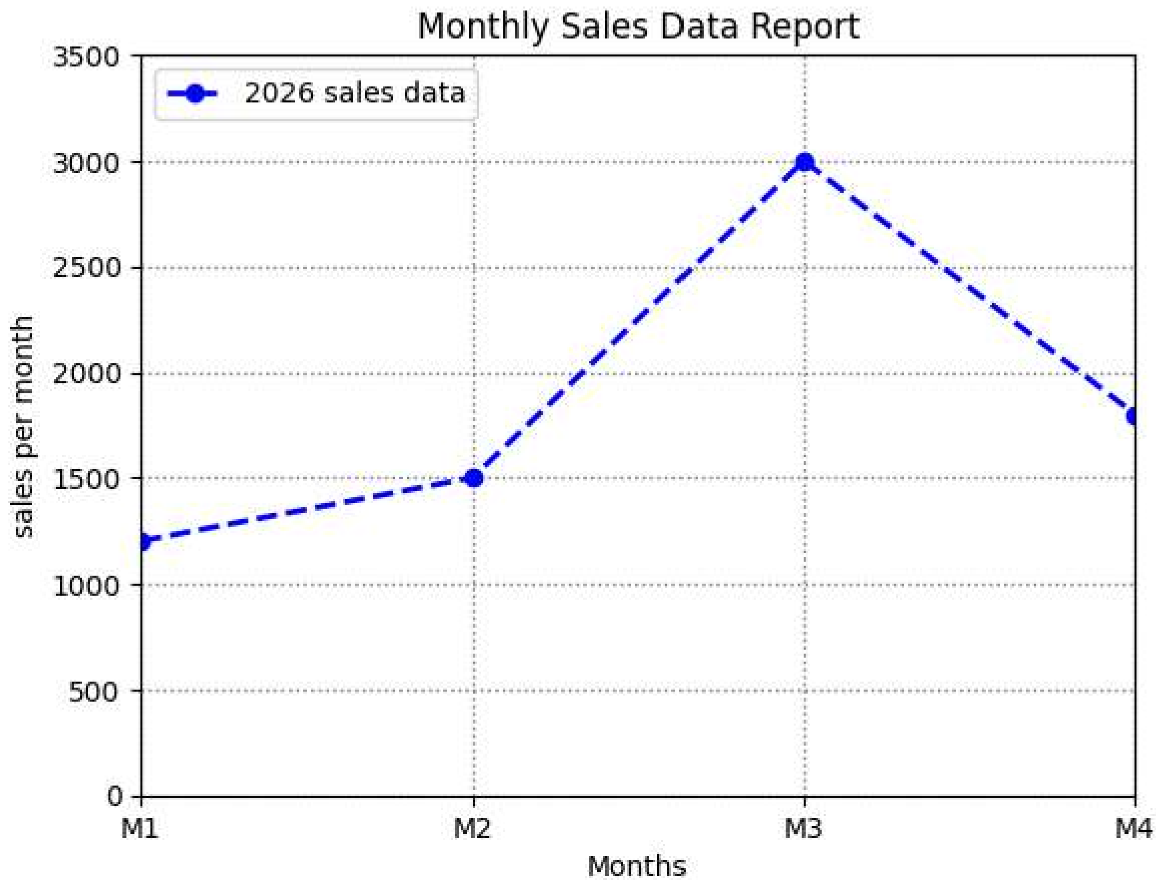
In [15]:
```
import matplotlib.pyplot as plt

data = {
    "Ali": 10,
    "Ahmed": 15,
    "Sara": 7,
    "Hira": 12
}

x = list(data.keys())
y = list(data.values())

plt.bar(x, y)
plt.show()
```



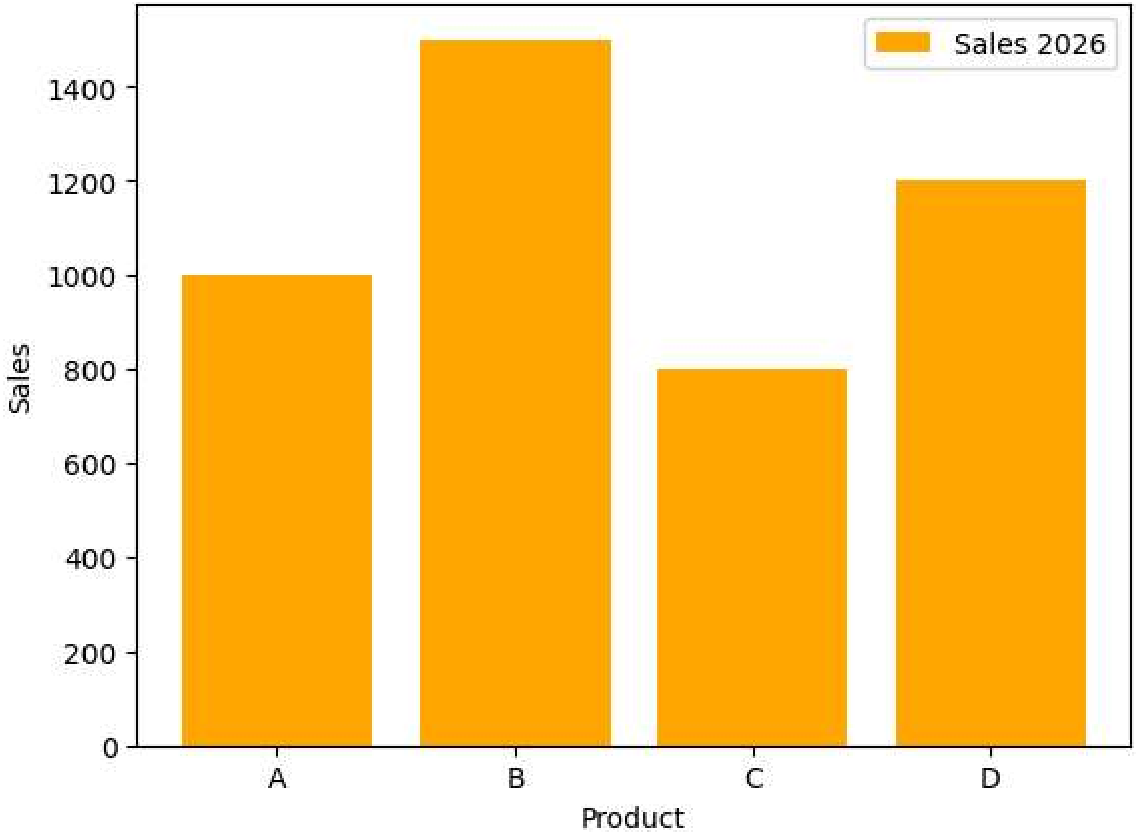
pyplot Functions

In [18]:

```
import matplotlib.pyplot as plt
x=[1,2,3,4]
y=[1200,1500,3000,1800]
# plt.plot(x,y,color="blue",linestyle="line_style",linewidth=value,marker="marker s
plt.plot(x,y,color="blue",linestyle="--",linewidth=2,marker="o",label="2026 sales d
plt.xlabel("Months")
plt.ylabel("sales per month")
plt.title("Monthly Sales Data Report")
plt.legend(loc="upper left")
plt.grid(color="gray",linestyle=":",linewidth=1)
plt.xlim(1,4)
plt.ylim(0,3500)
plt.xticks([1,2,3,4],["M1","M2","M3","M4"])
plt.show()
```

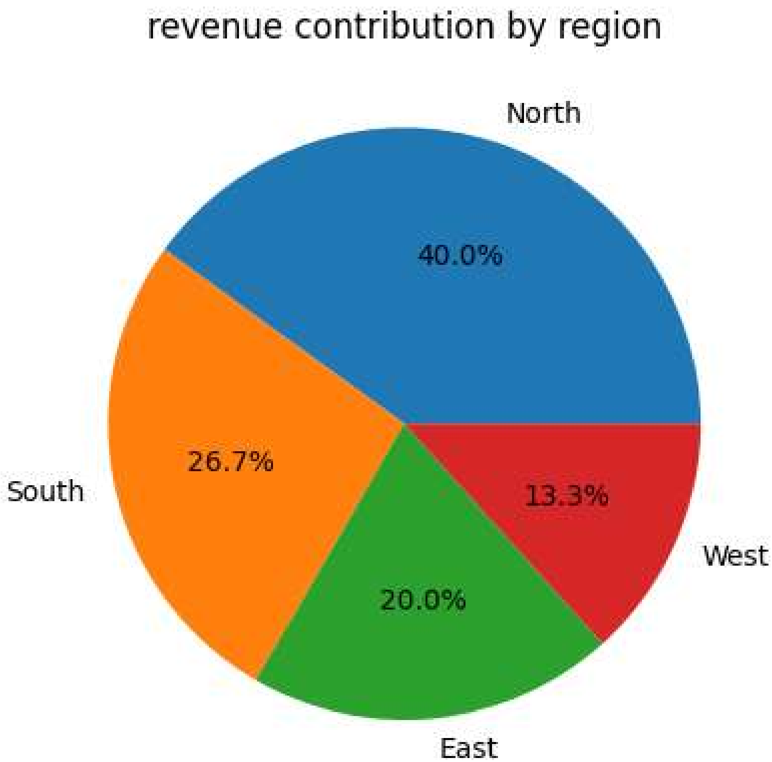


In [20]:
```
import matplotlib.pyplot as plt


# plt.bar(x,height,color="blue",width=value,label="label name")

product = ["A","B","C","D"]
sales = [1000,1500,800,1200]

plt.bar(product,sales,color="orange",label="Sales 2026")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.legend()
plt.show()
```

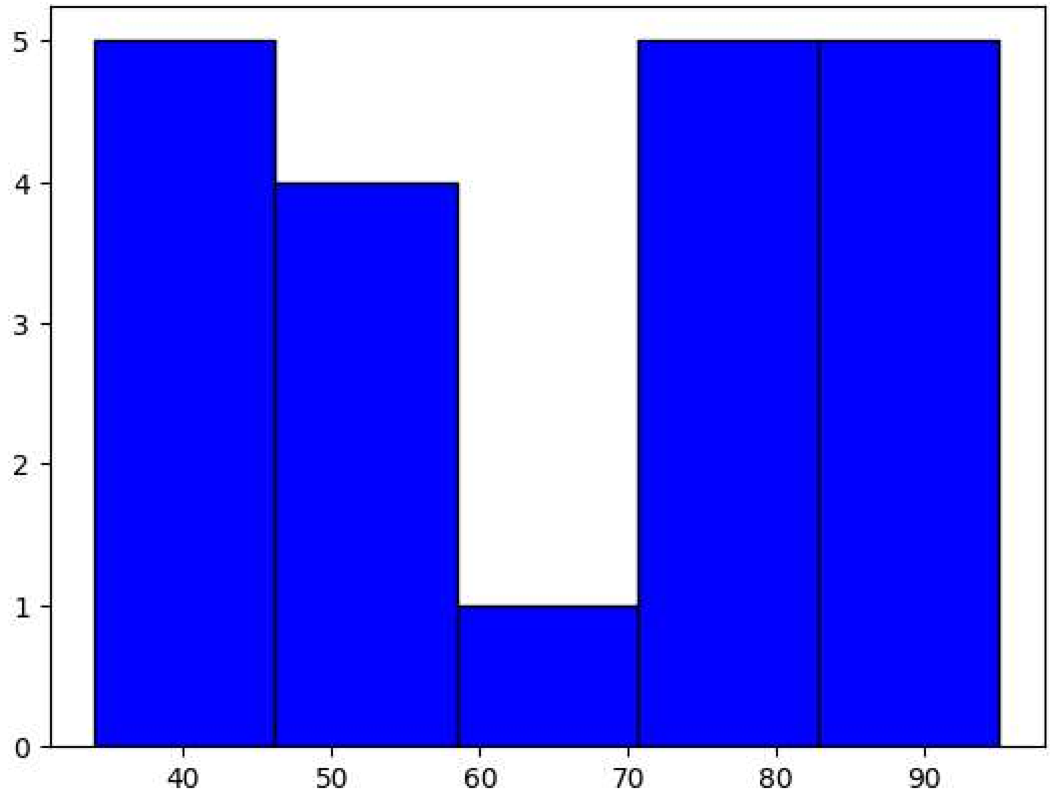


In [23]:

```
import matplotlib.pyplot as plt

# plt.pie(values,labels=label_list,colors=colors_list,autopct="%1.1f%%")

regions = ["North","South","East","West"]
revenue = [3000,2000,1500,1000]

plt.pie(revenue,labels=regions,autopct="%1.1f%%")
plt.title("revenue contribution by region")
plt.show()
```

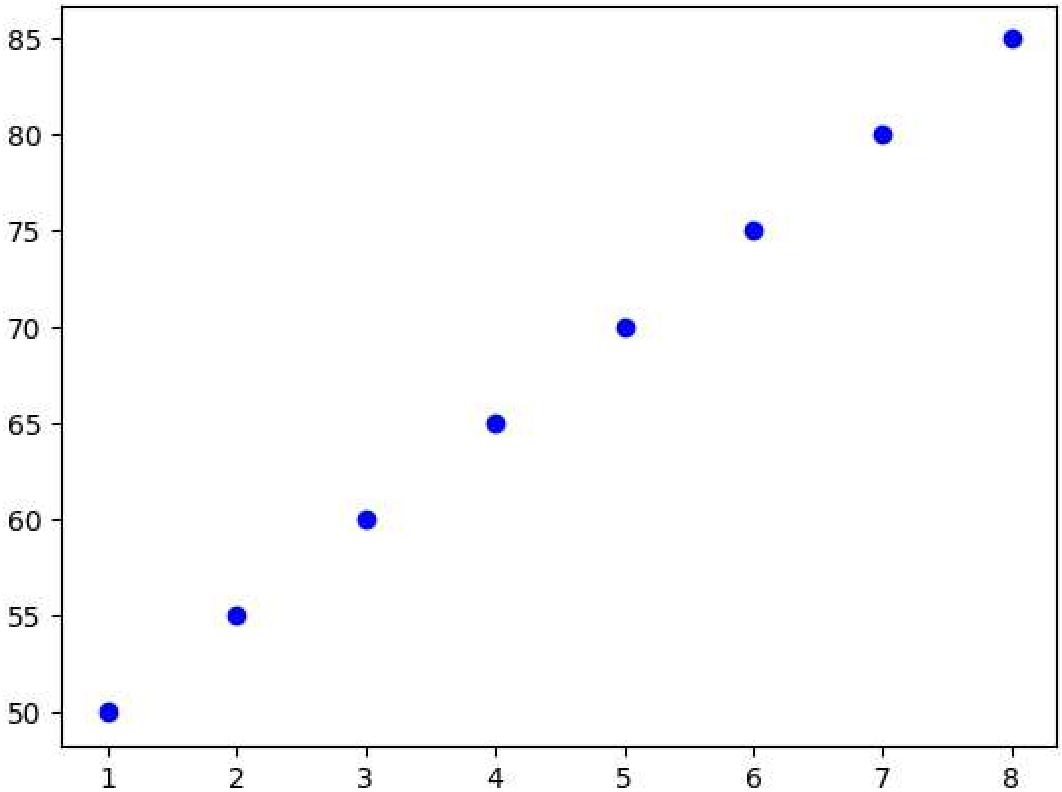


In [24]:
```
import matplotlib.pyplot as plt

# plt.hist(data,bins=number_of_bins,color="Color_name",edgecolor="black")

scores = [45,67,34,76,92,45,78,51,72,82,48,43,47,92,95,72,54,85,83,44]

plt.hist(scores,bins=5,color="blue",edgecolor="black")
plt.show()
```

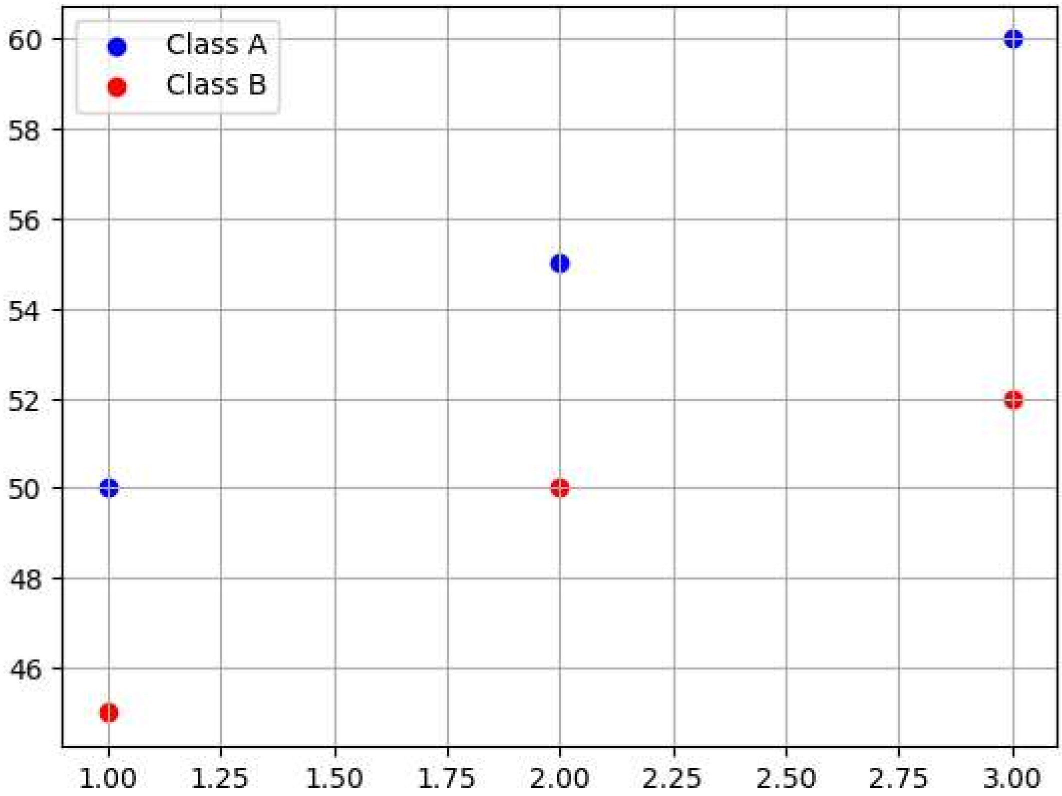


In [25]:

```
import matplotlib.pyplot as plt

# plt.scatter(x,y,color="blue",marker="marker symbol",label="label name)

hours_studies = [1,2,3,4,5,6,7,8]
exam_scores = [50,55,60,65,70,75,80,85]

plt.scatter(hours_studies,exam_scores,color="blue",marker="o",label="Student Data")
plt.show()
```



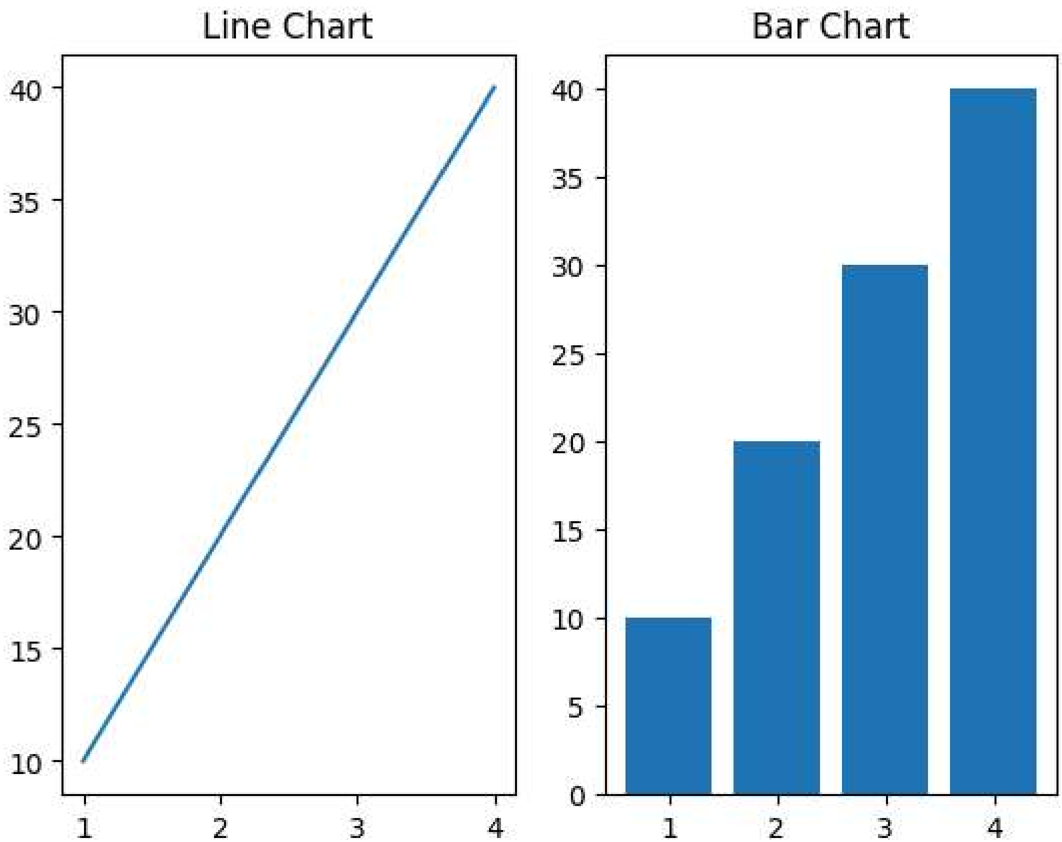
In [29]:
```
import matplotlib.pyplot as plt

plt.scatter([1,2,3],[50,55,60],color="blue",label="Class A")
plt.scatter([1,2,3],[45,50,52],color="red",label="Class B")
plt.grid()
plt.legend()
plt.show()
```



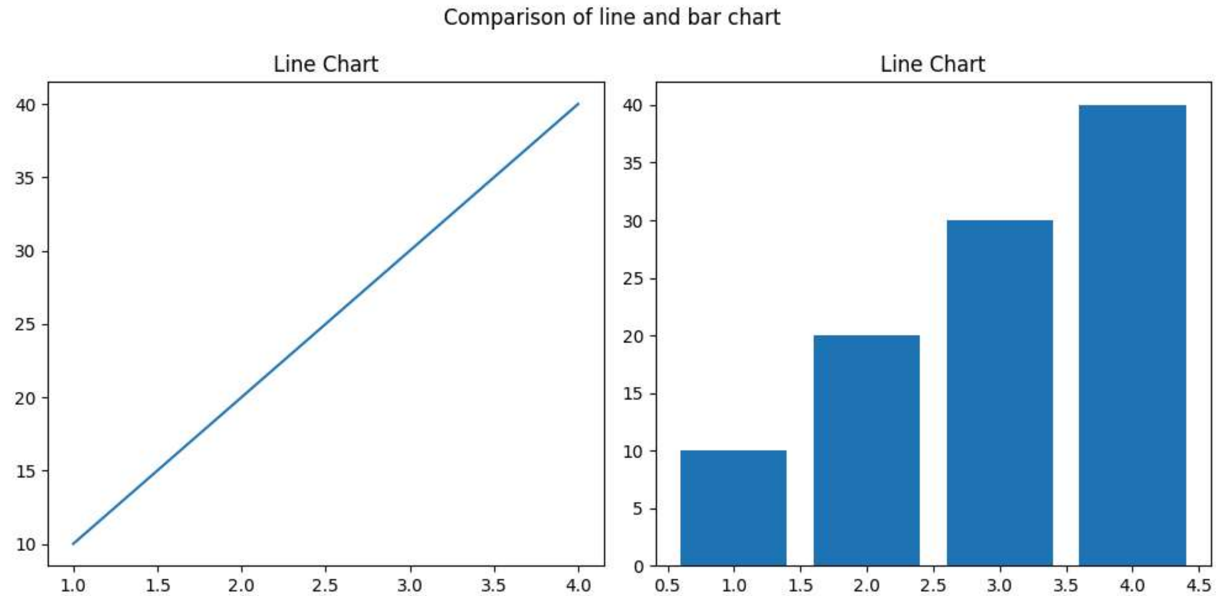
Subplot

In [37]:
```
import matplotlib.pyplot as plt

# plt.subplot(nrows,ncol,index)

x = [1,2,3,4]
y = [10,20,30,40]

plt.subplot(1,2,1)
plt.plot(x,y)
plt.title("Line Chart")

plt.subplot(1,2,2)
plt.bar(x,y)
plt.title("Bar Chart")
```

Out[37]:
```
Text(0.5, 1.0, 'Bar Chart')
```

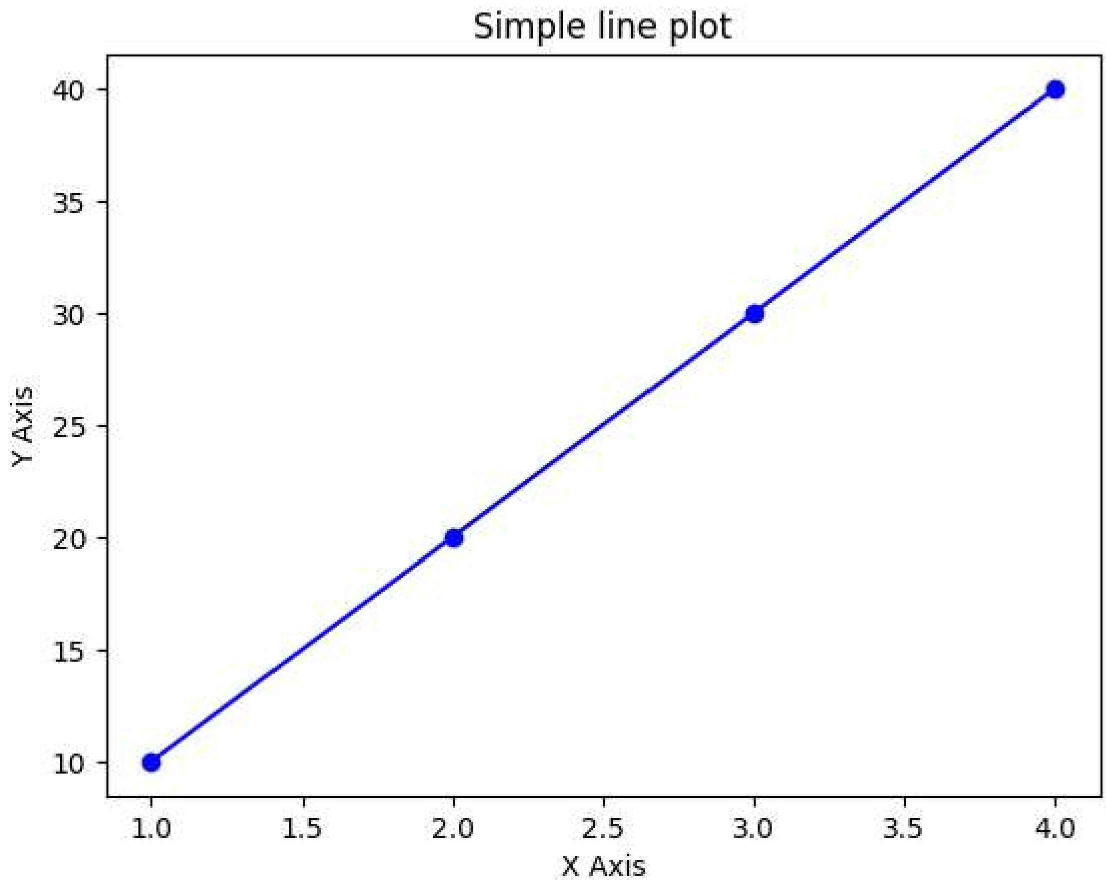


In [43]:
```
import matplotlib.pyplot as plt

# fig, ax = plt.subplot(nrows,ncol,figsize=(width,height))

x = [1,2,3,4]
y = [10,20,30,40]

fig, ax = plt.subplots(1,2,figsize=(10,5))

ax[0].plot(x,y)
ax[0].set_title("Line Chart")


ax[1].bar(x,y)
ax[1].set_title("Line Chart")

fig.suptitle("Comparison of line and bar chart")
plt.tight_layout() # automatic layout ko adjust kr data hy
# ya show() say pahla wali line ma use hota hy
plt.show()
```



Saving Figures

In [44]:
```
import matplotlib.pyplot as plt

# savefig("filename.extension",dpi=value,bbox_inches="tight")

x = [1,2,3,4]
y = [10,20,30,40]

plt.plot(x,y,color="blue",marker="o")
plt.title("Simple line plot")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")

plt.savefig("line_chart.png",dpi=300,bbox_inches="tight")
plt.show()
```


Simple line plot
40
35
30
25
20
15
10
Y Axis
1.0
1.5
2.0
2.5
3.0
3.5
4.0
X Axis